Appendix E. Installing MySQL and creating a database

In this appendix, I will show you how to create a MySQL database. In some of the examples we implement in chapters 12 through 14, we use an external database management system (DBMS). For these examples, you'll need to create a database for your app to use prior to implementing the project.

For the database technology, you have plenty of options to choose from, like MySQL, Postgres, Oracle, MS SQL server, and so on. I encourage you to select the technology you like if you already have a preferred one. For the examples in the book, I had to choose a certain database technology, and I decided to use MySQL, which is free, lite, and easy to install on any operating system. You'll find MySQL used in general for tutorials and examples.

For learning Java and Spring, it's irrelevant which DBMS technology you use. The Java and Spring classes and methods are the same, whether you choose MySQL, Oracle, Postgres, or any other relational database technology.

The steps for creating a database for use in the examples are the following:

1. Install the DBMS on your local system; we will use MySQL.
2. Install a client application for the DBMS; we will use MySQL Workbench, which is one of the most known client apps for MySQL.
3. In the client app, connect to the local DBMS installation.
4. Create the database you want to use in the example.

Step 1: Install a DBMS on your local system

The first step is to make sure you have a DBMS to work with. The examples in this book use MySQL, but you can choose to install another DBMS if you prefer using another technology. If you choose to install MySQL, you can find the installer for download here: **https://dev.mysql.com/downloads/mysql/**.

Download the installer according to your operating system and follow the steps as presented in the installation guide here: https://dev.mysql.com/doc/refman/8.0/en/installing.html.

Note that during the installation of the DBMS you might be required to create an account. Remember these credentials because you'll need them in step 3.

Step 2: Install a client application for your DBMS

To work with the DBMS, you need a client app, which is used to create the database and sometimes alter its structure and validate your app. You need to install a client app according to the DBMS technology of your choice. If you work with MySQL, you can use MySQL Workbench, one of the most used MySQL clients.

Download MySQL Workbench installer according to your operating system (https://dev.mysql.com/downloads/workbench/) and install it as described in the installation manual: https://dev.mysql.com/doc/workbench/en/wb-installing.html.

Step 3: Connecting to the local DBMS

Once you install the DBMS on your local system and a client for your DBMS, you need to connect the client to the DBMS. To set up the connection, you need the credentials you configured in step 1. If the DBMS installation didn't request you set up a username and password, you could use the username "root" with an empty password.

I show you how to add the connection in MySQL Workbench. As presented in figure E.1, once you open MySQL Workbench, click the plus icon near "MySQL Connections."

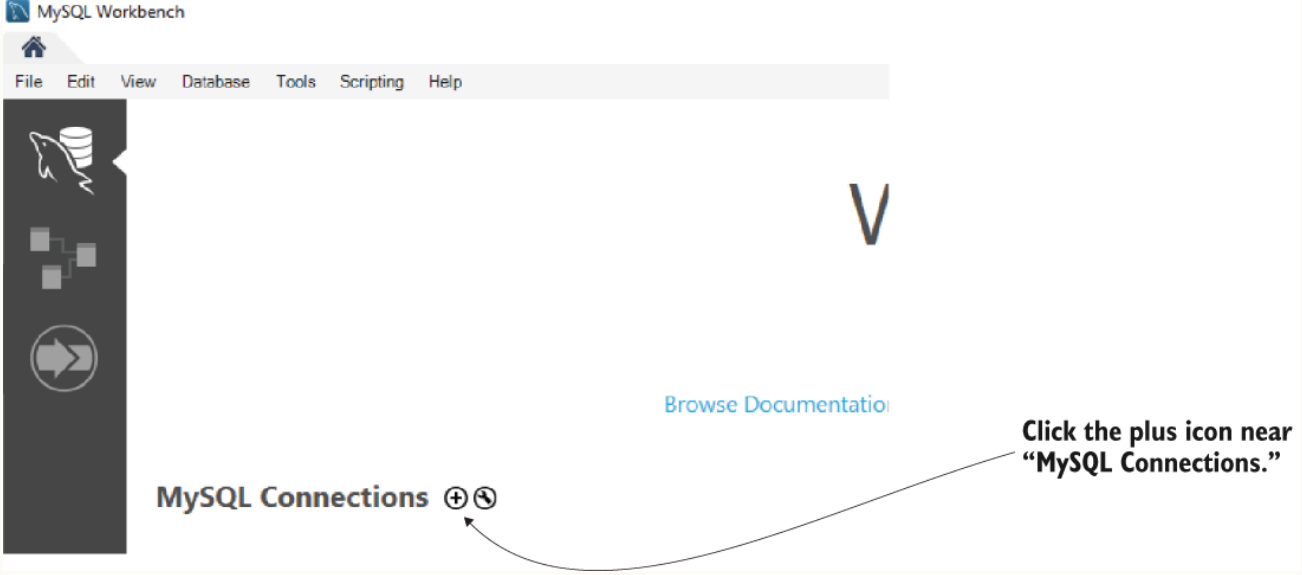


Figure E.1 Once you open MySQL Workbench, click on the plus icon near the label "MySQL Connections" to add a new connection.

A popup window appears once you click the plus button, and you have to give the connection a name and use the credentials you configured during the DMBS installation. MySQL Workbench uses these credentials to authenticate at the DBMS (figure E.2).

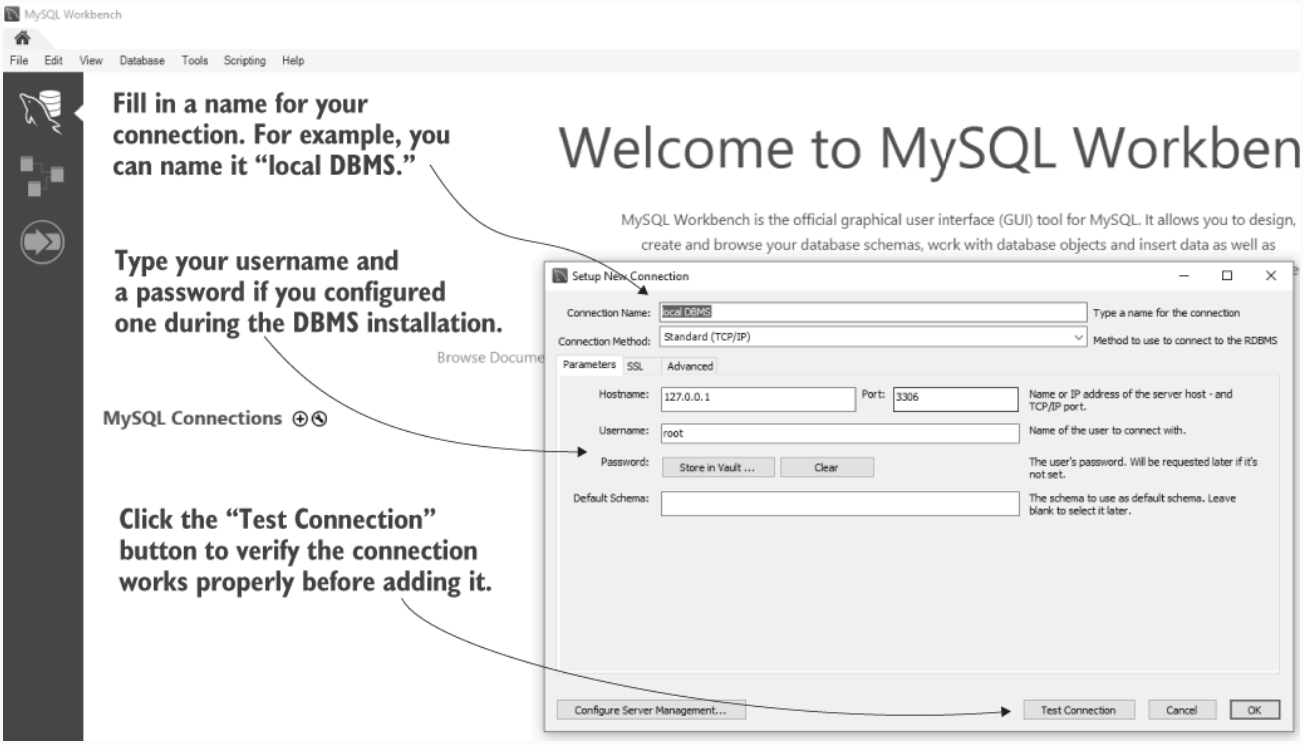


Figure E.2 Give the connection a name and fill in the credentials you configured when installing the DBMS. Then, click on the "Test Connection" button to verify that MySQL Workbench can connect to the database.

For some MySQL versions, MySQL Workbench might display a warning message. For implementing the examples in this book, this won't affect you, so click the "Continue Anyway" button if you get such a message (figure E.3).

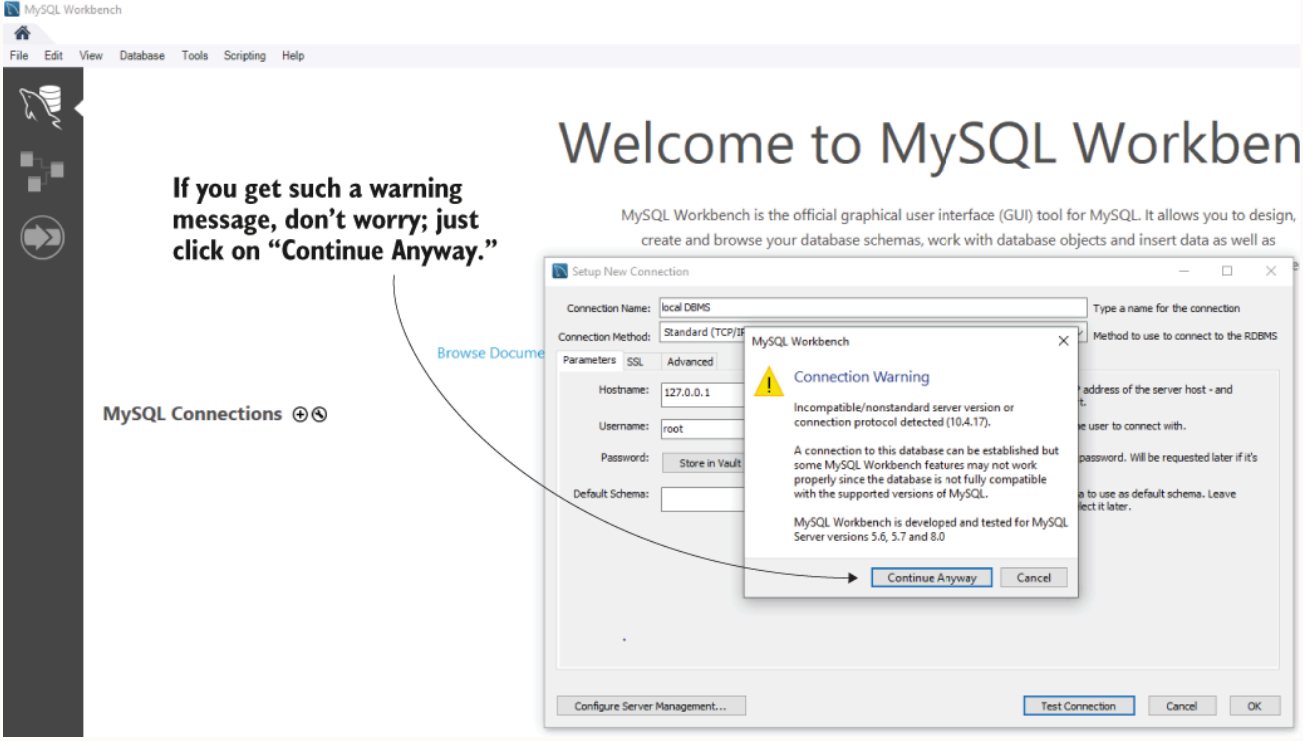


Figure E.3 Sometimes MySQL Workbench displays a warning message. This warning won't affect the way you implement the examples in the book. You can click on the "Continue Anyway" button if such a message is displayed.

If the connection details are correct and MySQL Workbench managed to connect to your local DBMS, you will see a popup like the one in figure E.4.

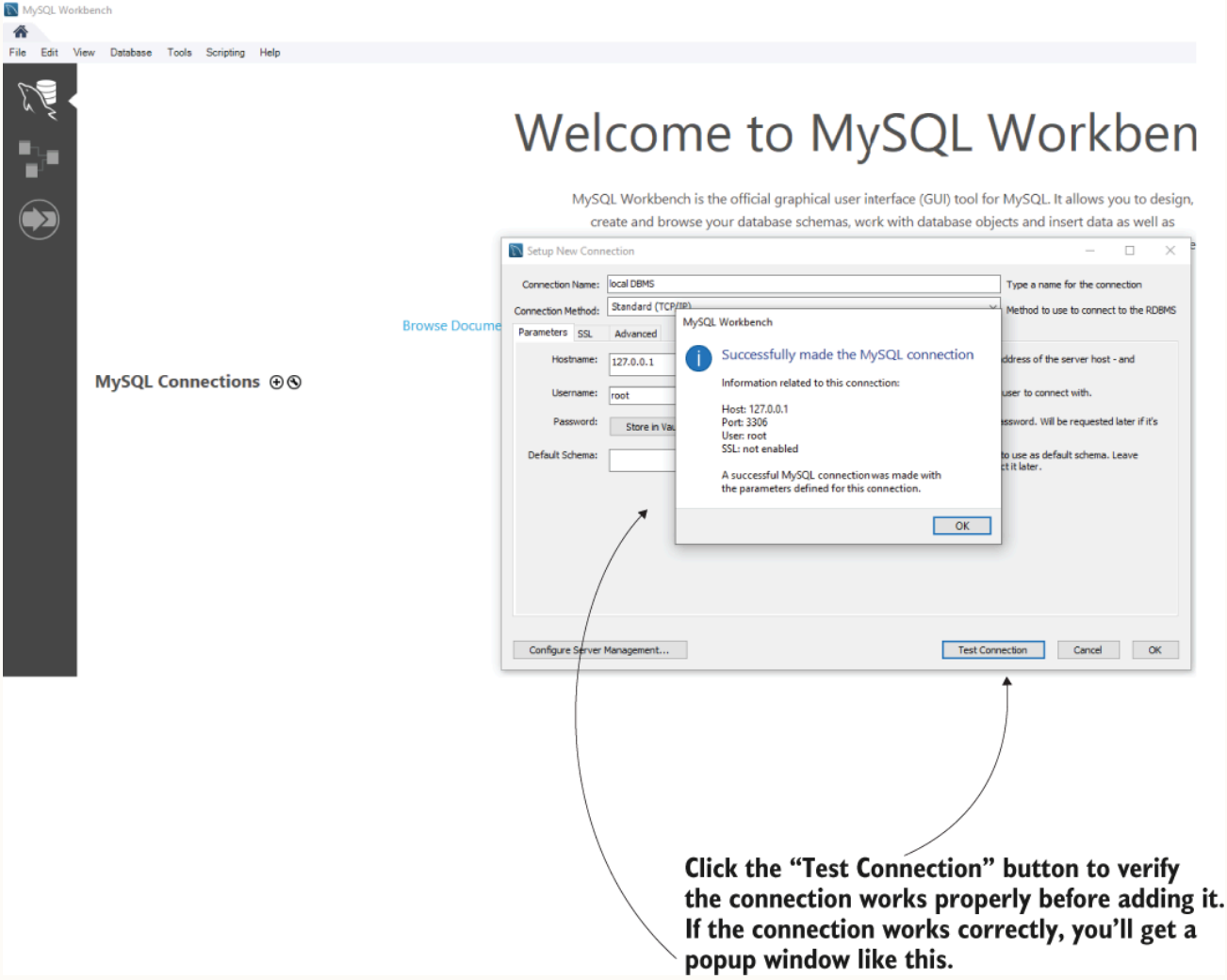


Figure E.4 If MySQL Workbench connected to the local DBMS, it displays a "Successfully made the MySQL connection" message in a popup dialog.

Once you click the OK button, the new connection is added, and in the MySQL Workbench home screen, you can now see it as a rectangle with the name you gave it (figure E.5).

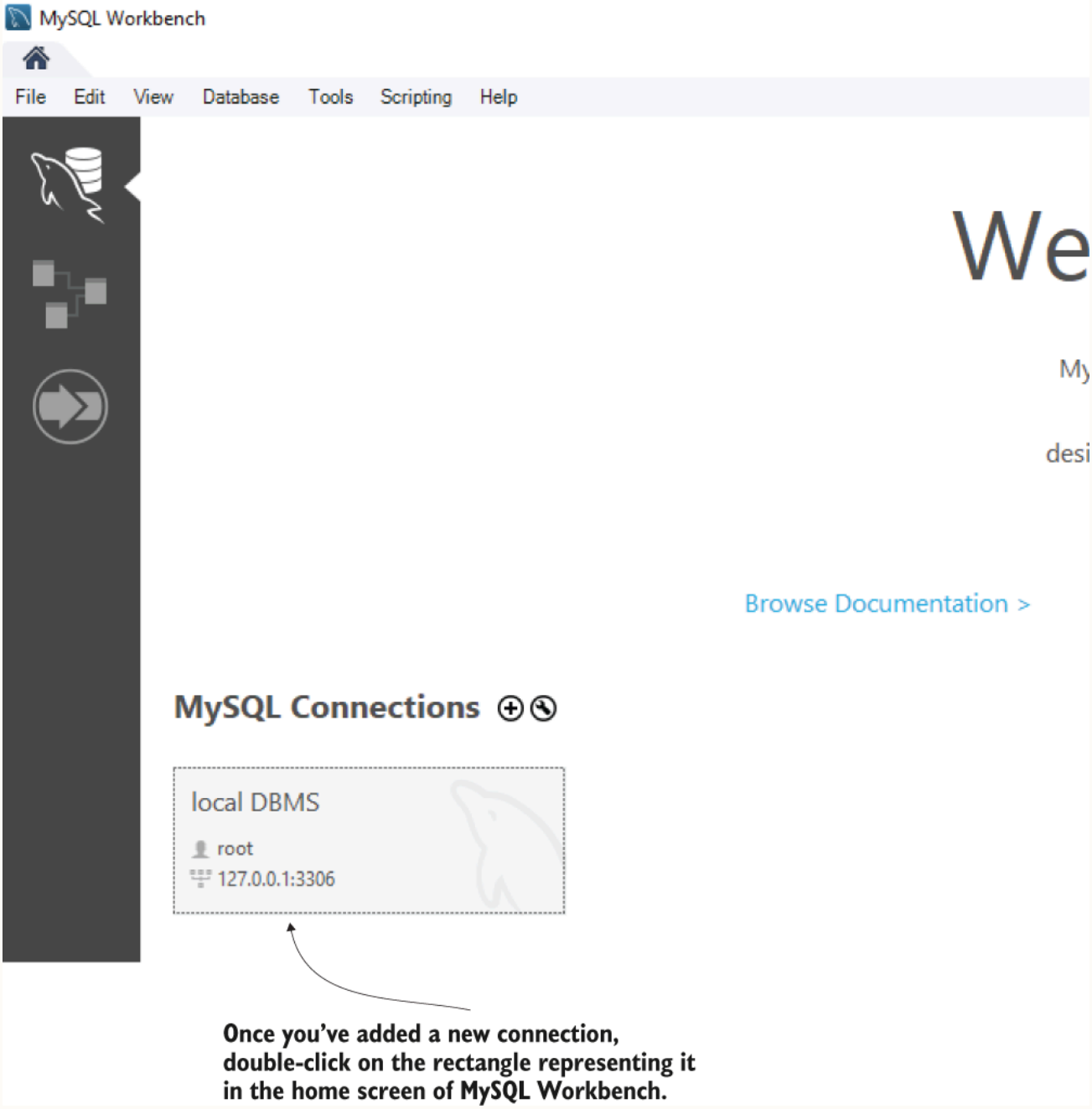


Figure E.5 In the MySQL Workbench main screen, you can now see the connection you added. The connection is displayed as a gray rectangle, and it has the connection's name.

Step 4: Add a new database

Now that you've added a connection, it's time to use it and create a database. The examples we implement in chapters 12 through 14 use databases. You need to add a database before developing an app that uses it.

After you double click on the rectangle in the MySQL Workbench home screen that represents the connection to the local DBMS, you will see a screen like the one in figure E.6. Click on the small cylinder icon in the toolbar to add a new database.

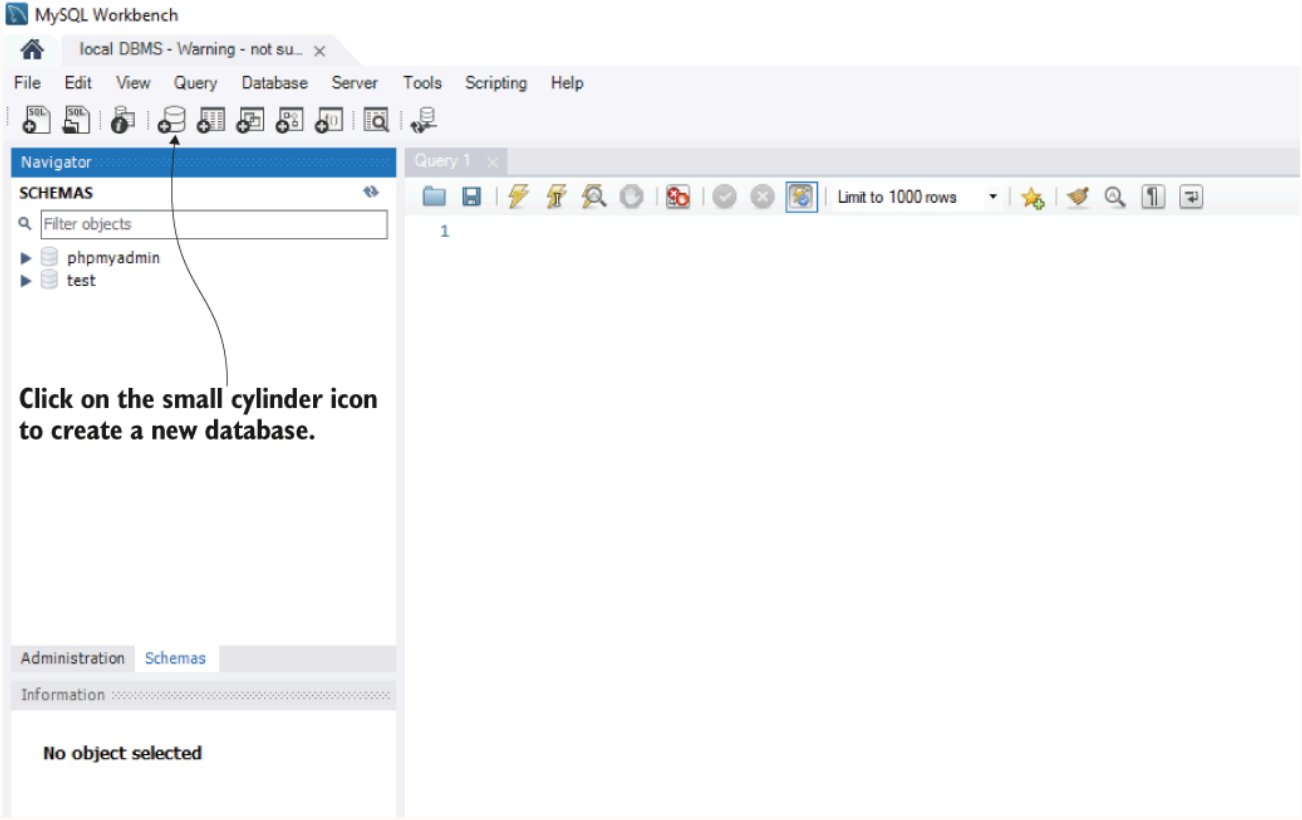


Figure E.6 After connecting to the DBMS, you can add a new database by clicking on the toolbar's small cylinder icon.

You have to give the database a name, then click on the Apply button, as presented in figure E.7.

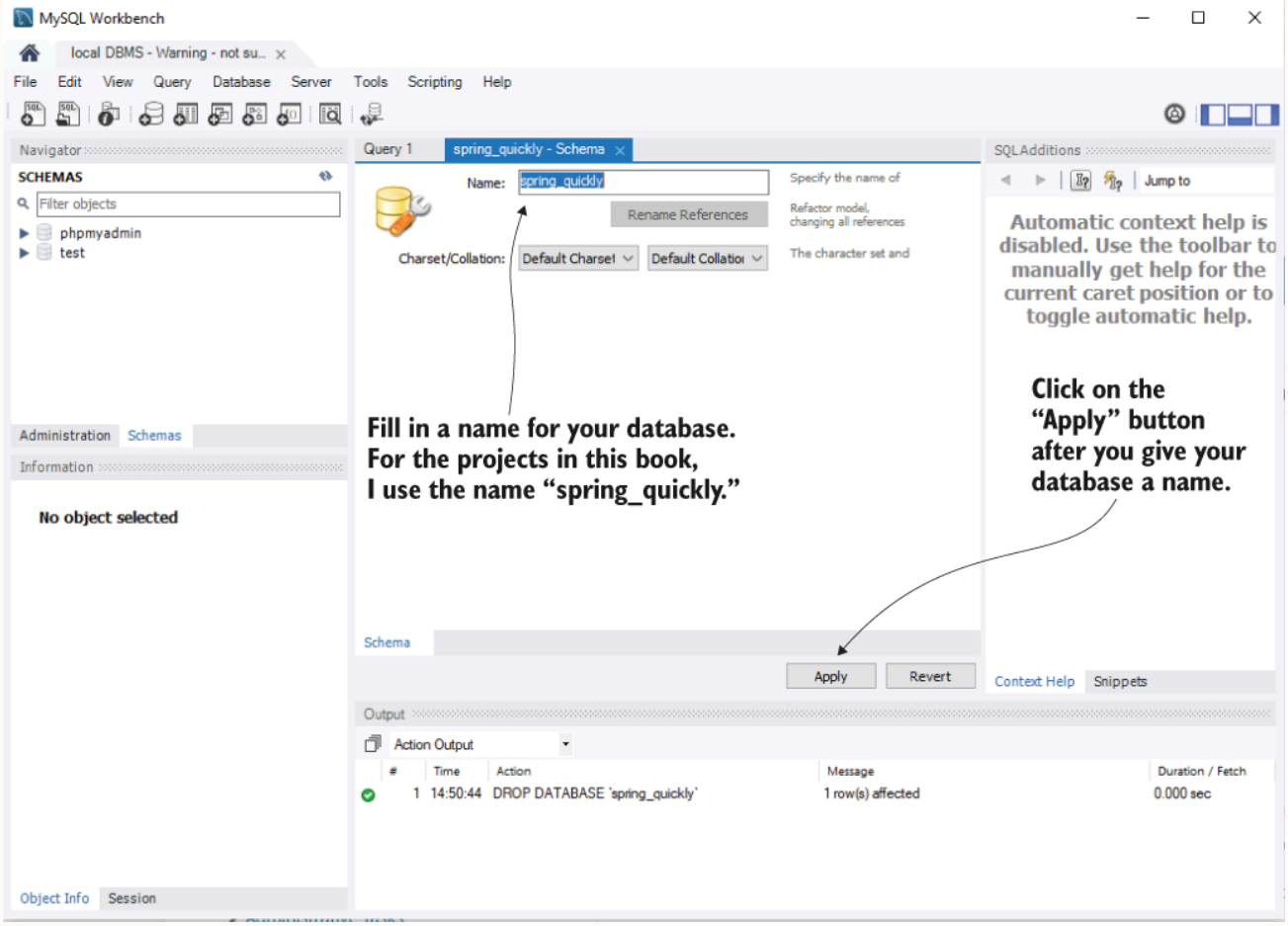


Figure E.7 Give the database a name. Our examples use "spring_quickly." Click on the Apply button to create the database.

Before adding the database, MySQL Workbench asks you to confirm once more. Click the Apply button again, as presented in figure E.8.

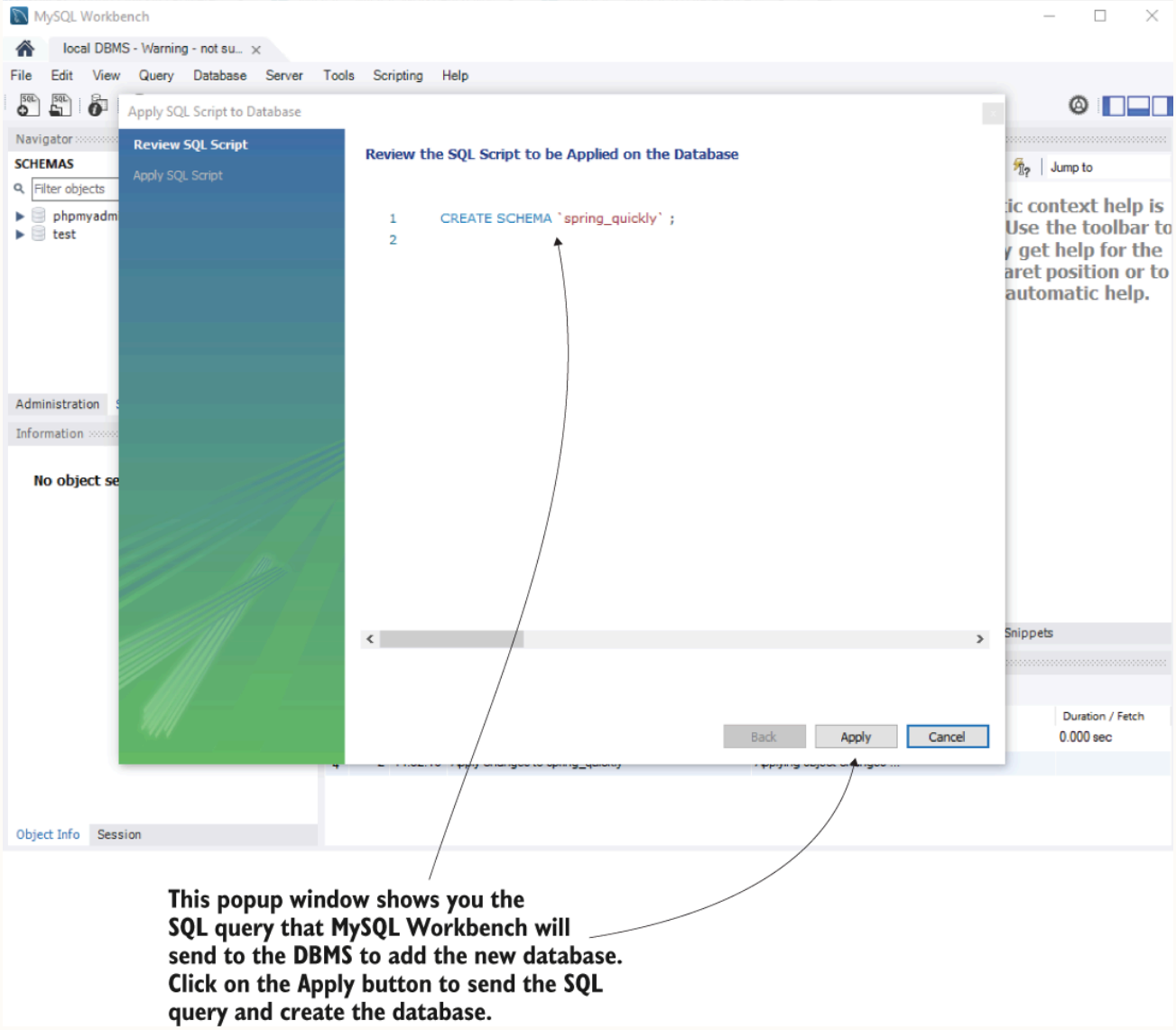


Figure E.8 You need to again confirm the creation of the database. In this screen, you also see the query MySQL Workbench sends to the DBMS to create the new database. Click the Apply button to execute the query and create the new database.

If you find your new database on the left side of the window, it means you successfully created it, and your Spring app should be able to use it as well (figure E.9).

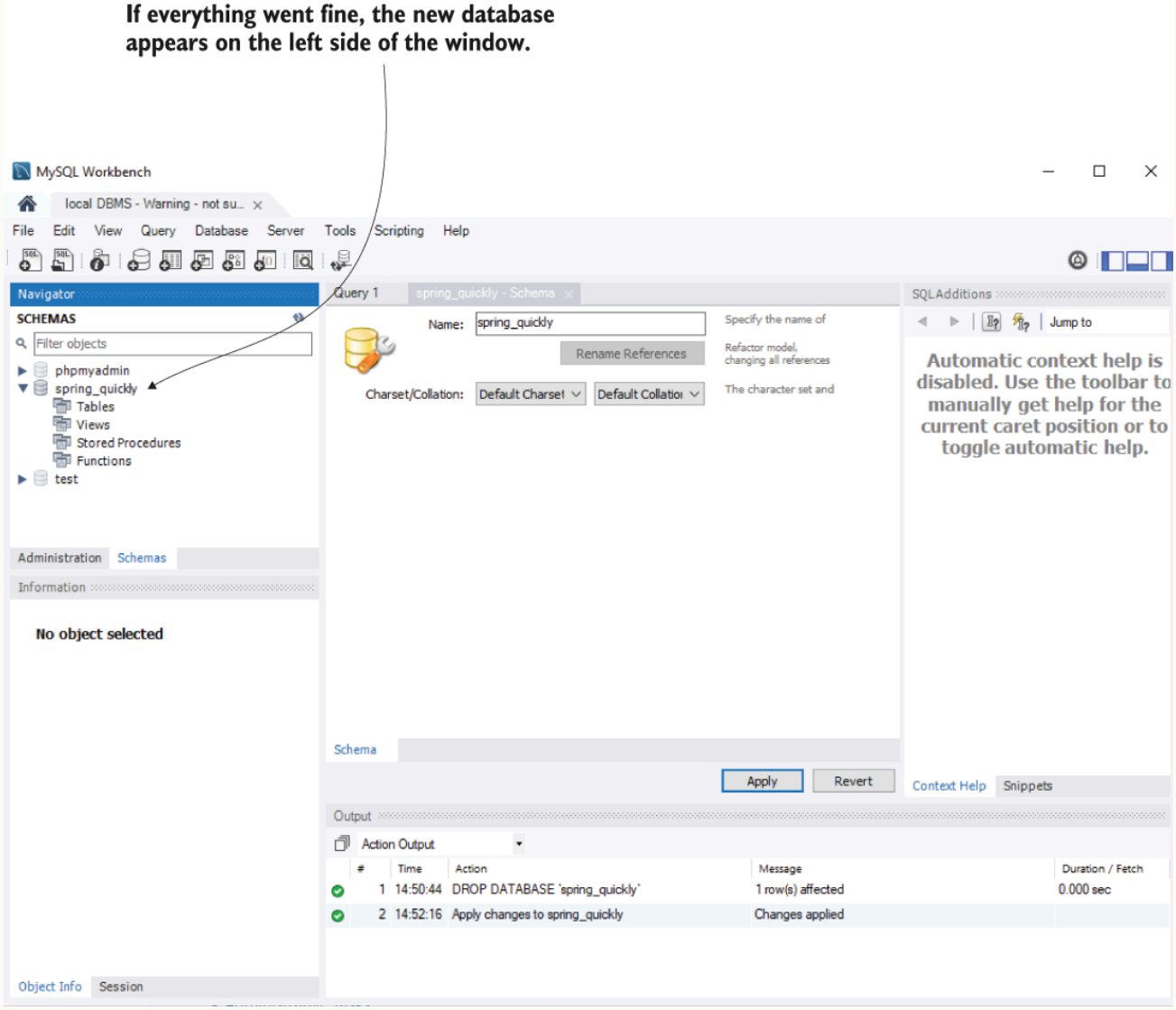


Figure E.9 The new database appears on the left side of the window. If you can see it there, it means you successfully created a new database.

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